

Alessandro Pani Patiño

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Computer Science student at Universidad de Costa Rica (B.Sc. expected 2027) with independent research in machine learning and combinatorics alongside a strong full-stack engineering background. I design and run large empirical ML studies (13k+ experiments), build production systems end to end (React, PHP/Oracle, Node.js), and turn open research questions into reproducible, well-engineered code. Seeking a software engineering or machine learning internship.

Core Competencies

- **ML & Research:** scikit-learn, NumPy, pandas, PyTorch; experiment design, model evaluation, effective-complexity measures, LLM integration & RAG, prompt engineering
- **Languages:** Python, TypeScript, JavaScript, PHP, Dart, Java, C++, SQL
- **Front-end:** React 19, Next.js, Svelte, Flutter, Material UI, Vite
- **Back-end & APIs:** Node.js, PHP 8.4, Django, FastAPI, REST API design, WebSockets, WebRTC
- **Databases:** Oracle (PDO/OCI), PostgreSQL, SQL, Supabase, MongoDB
- **Tooling & Practices:** Docker, Git, CI/CD, PHPStan, Linux, layered architecture, unit testing, Agile

Professional Experience

- **Universidad de Costa Rica, Vicerrectoría de Administración** **San José, CR**
Full-Stack Developer – SIGECAT *2026–Present*
 - Built, as part of an academic assistance program, a full-stack workload & work-shift declaration platform: employees declare shift functions, licenses and rest periods while administrators manage the university's organizational hierarchy (areas, departments, sections, units, positions).
 - Developed the front-end as a React 19 + TypeScript SPA (Vite, Material UI) with client-side PDF export of declarations.
 - Designed a framework-less REST API in PHP 8.4 on Oracle Database using a clean layered architecture: thin controllers, a service layer, and PDO/OCI repositories with prepared statements.
 - Engineered token-based auth with refresh-token rotation, plus bcrypt hashing, per-IP rate limiting and automatic account lockout.
 - Set up a Dockerized monorepo with idempotent SQL migrations and a CI pipeline running lint, build and PHPStan static analysis on every pull request.
- **ECCI Control – Universidad de Costa Rica** **San José, CR**
Software Engineer *2024–2025*
 - Built a QR-based electronic access-control system for an academic assistance program, replacing manual sign-in with a secure, traceable digital process.
 - Gathered requirements through interviews with students and faculty and delivered the deployed system.
- **deferia.cr (UCR x FAO)** **Remote**
Software Engineer *2023–2025*
 - Built a data-driven platform on CNP agricultural price data using Django and PostgreSQL.
 - Cleaned and normalized datasets and implemented interactive visualizations for historical price analysis.
- **Independent** **Costa Rica / Remote**
Freelance Web Developer *2022–Present*
 - Designed, built and shipped websites for 10+ clients across legal (VM Legal Group), hospitality and e-commerce, mostly React-based.
 - Own the full delivery cycle: requirements, development, deployment, SEO and analytics; maintain a production hotel site (issimosuites.com).

Research Projects

- **Python, scikit-learn, NumPy** 2026
Feature-wise Double Descent – empirical study (draft paper)
 - Ran 13,000+ training runs mapping the feature-wise double-descent error curve across 4 dimensionality reducers, 7 model families and 5 datasets on a single 4-core CPU.
 - Showed the error peak is a spectral phenomenon — locatable without training from the reduced design's smallest singular value — and identified two distinct peak regimes; built a resumable sweep harness with effective-degrees-of-freedom estimation (Ye, 1998).
- **Python, SQLite, NetworkX, nauty** 2026
Conjecture Engine – automated graph-theory conjecturing
 - Built a Graffiti/TxGraffiti-style system that runs unattended overnight: enumerates graphs, computes invariants into SQLite, and proposes then refutes inequality conjectures with an LLM advisor in the loop.
 - Accumulated 200+ surviving conjectures, automatically rediscovering classical results (e.g. Whitney's $\kappa \leq \lambda_e \leq \delta$) as sanity checks.

Software Projects

- **Svelte, WebRTC, WebSockets, Node.js** Live
MyVRRoom – Real-time video platform
 - Built a peer-to-peer video platform on a secure WebRTC mesh with WebSocket signaling, strict room isolation and anti-spoof validation.
 - Added synchronized shared features: a multiplayer game and a collaborative whiteboard.
- **Flutter, Supabase** Live
Mejengas – Real-time team sessions
 - Developed a Kahoot-style team-management app with anonymous session joining via link, code or QR (no account needed), using Supabase for real-time session state.
- **JavaScript, Canvas, WebGL, MediaPipe** Live
Museo del Espacio Matemático
 - Built 30+ interactive generative-art visualizations of math and physics (orbits, Fibonacci, fields), controllable by webcam hand-tracking, packaged as an installable PWA.
- **Python, Streamlit** Live
Geopolitical Risk Analytics
 - Interactive dashboard analyzing geopolitical risk indicators across countries, with dynamic filters and time-based visualizations.

Education

- **Universidad de Costa Rica (UCR)** San José, CR
B.Sc. Computer Science (Software Engineering) 2027 (expected)

Languages

- Spanish (native), Italian (native), English (professional working proficiency).